Principles of Spatial Analysis

Week 07: Geodemographics



UCL

This week

- Geodemographic classification.
- k-means clustering.

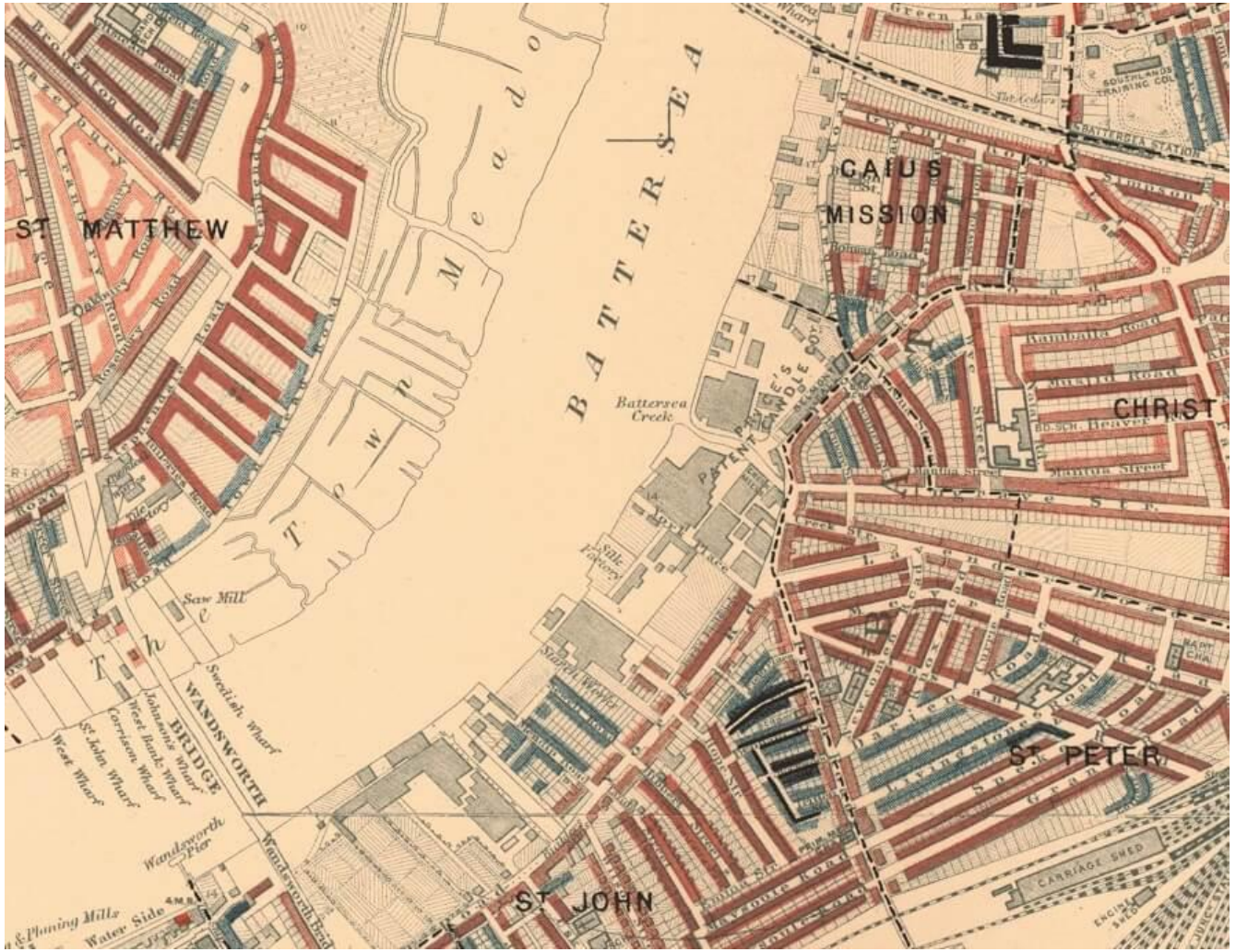
Geodemographics

- Analysis of people (*demographics*) by where they live (*geo*).
- Used to identify similar neighbourhoods or administrative areas.
- Means of multivariate data reduction for the differentiation of areas.
- Been around for many many years, arguably dating back to the late 1800s.

Charles Booth

- Created the first geodemographic-style classification.
- Shipping business owner and philanthropist.
- Survey: *Life and Labour of People in London*.
- He noticed that there is a geographical pattern in the distribution of different social categories: people who live in a particular neighbourhood and share similar living conditions, are of similar characteristics and social status.

Charles Booth



Maps Descriptive of London Poverty 1889. Charles Booth's *Inquiry into Life and Labour in London (1886–1903)*.

Charles Booth

Classification	Colour	
Lowest class. Vicious, semi-criminal.	Black	
Very poor, casual. Chronic want.	Dark blue	
Poor. 18s. to 21s. a week for a moderate family.	Light blue	
Mixed. Some comfortable others poor.	Purple	
Fairly comfortable. Good ordinary earnings.	Pink	
Middle class. Well-to-do.	Red	
Upper-middle and upper classes. Wealthy.	Yellow	

Geodemographics

- Further developed in 1970s to target urban deprivation funding.
- Commercial sector also got involved (CACI: ACORN / Experian: MOSAIC).
- Office for National Statistics' Output Area Classification (2001, 2011, 2021).
- ONS' Output Area Classification completely open using Census data.

Geodemographics

Singleton *et al*. 2020:

“A geodemographic classification is created by assembling a wide range of measures that describe the characteristics of areas and/or those people living within them, and then, through the implementation of unsupervised learning (clustering), identifies groups of areas that share common characteristics. Emerging clusters may be divided or aggregated to create a hierarchy, and it is typical that these be accompanied by labels, descriptions, photographs, diagrams and graphs.”

Variable selection

Choice of measures can create bespoke classifications:

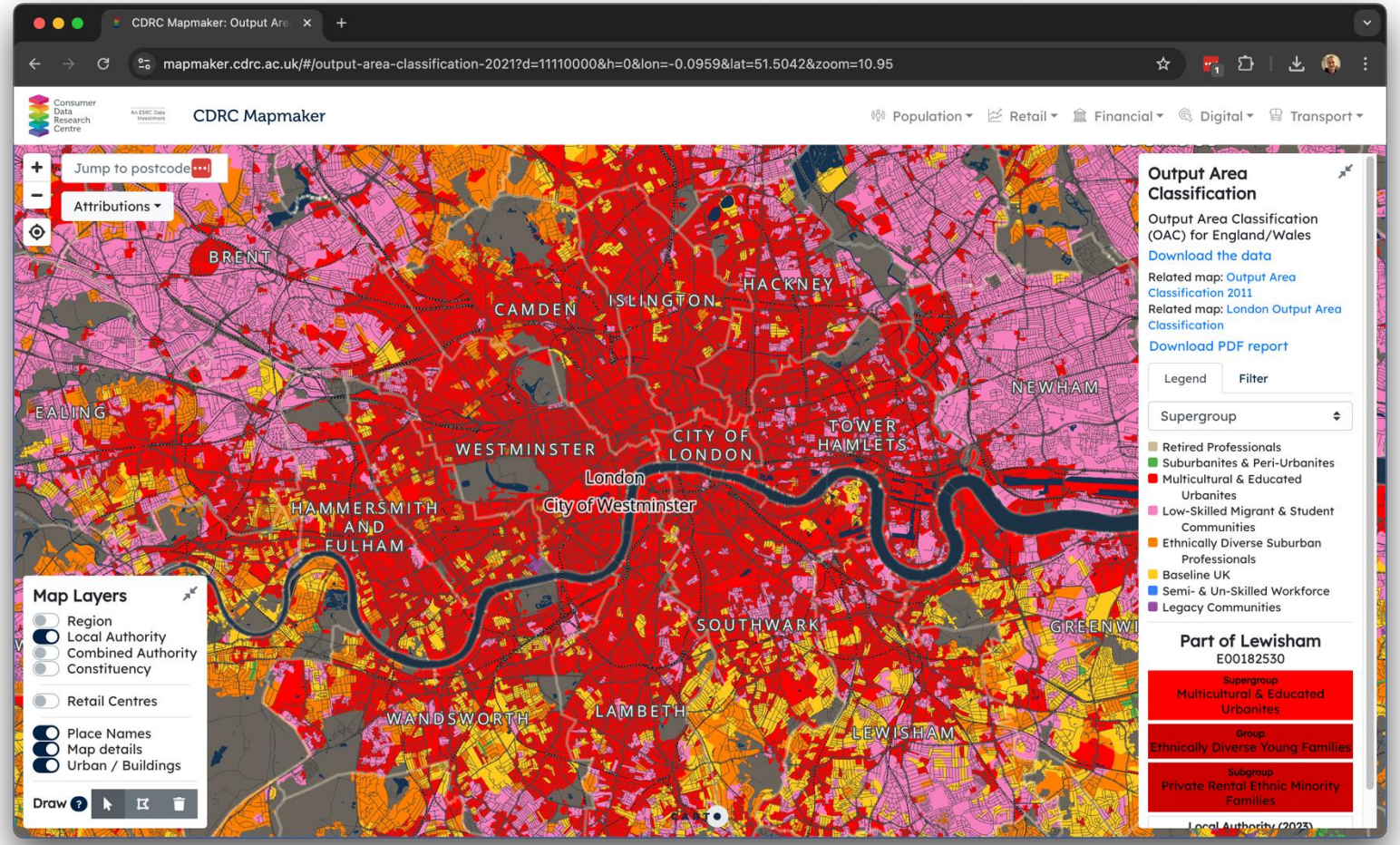
- Output Area Classification: ~60 census variables, focused on holistic understanding of area characteristics.
- Workplace Zones Classification: 48 census variables across 4 domains, focused on workers and workplaces.
- Internet User Classification: built from a range of consumer, survey and open (census) data focusing on how populations interact with the internet (use and engagement).

Clustering

Areas are grouped into clusters based on their variables and then mapped to their areal unit, e.g. Output Area, Workplace Zone:

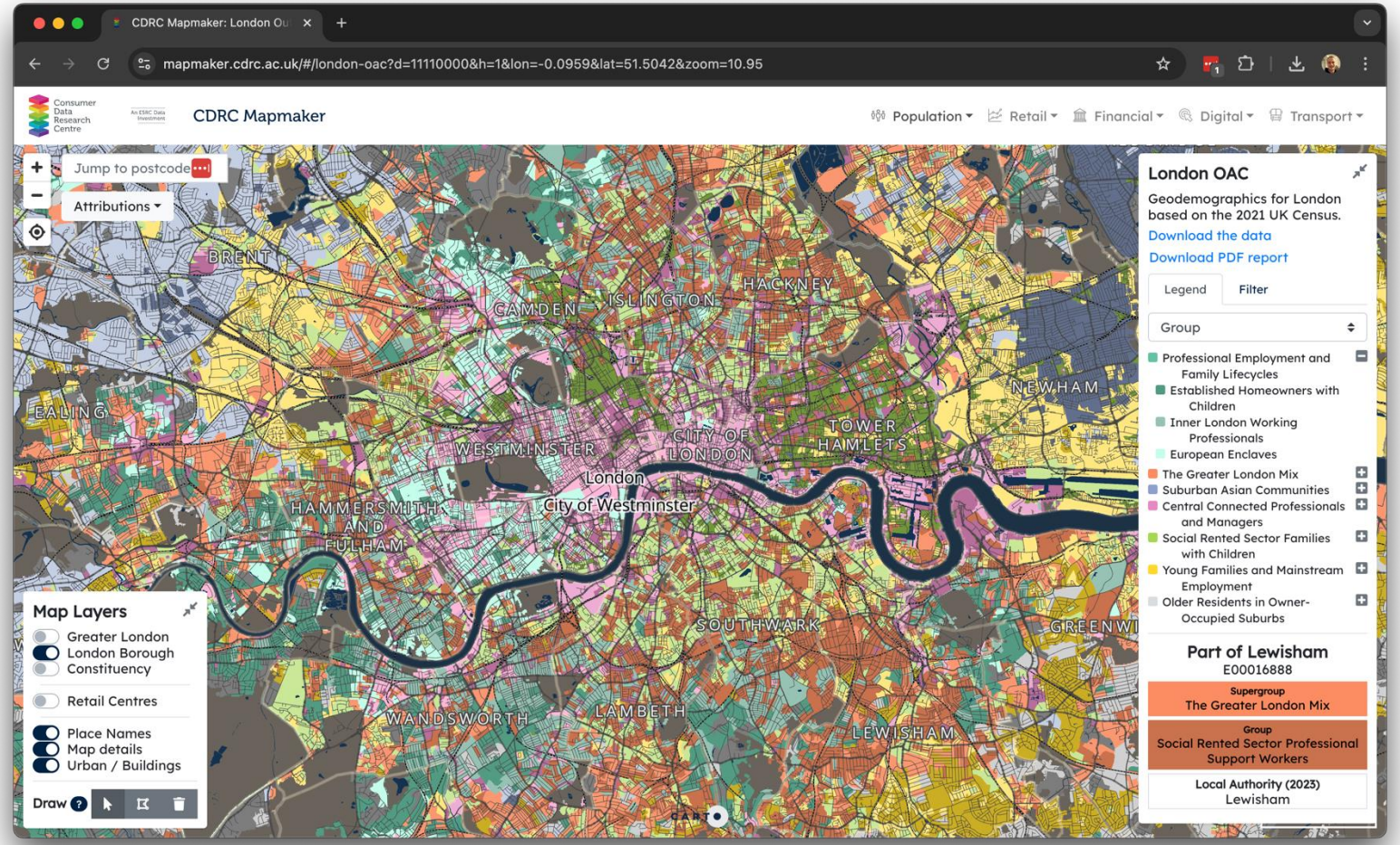
- Sometimes there is a hierarchy of clusters (e.g. supergroups, clustergroups).
- Each cluster will have a name and description (pen portrait).

Geodemographics



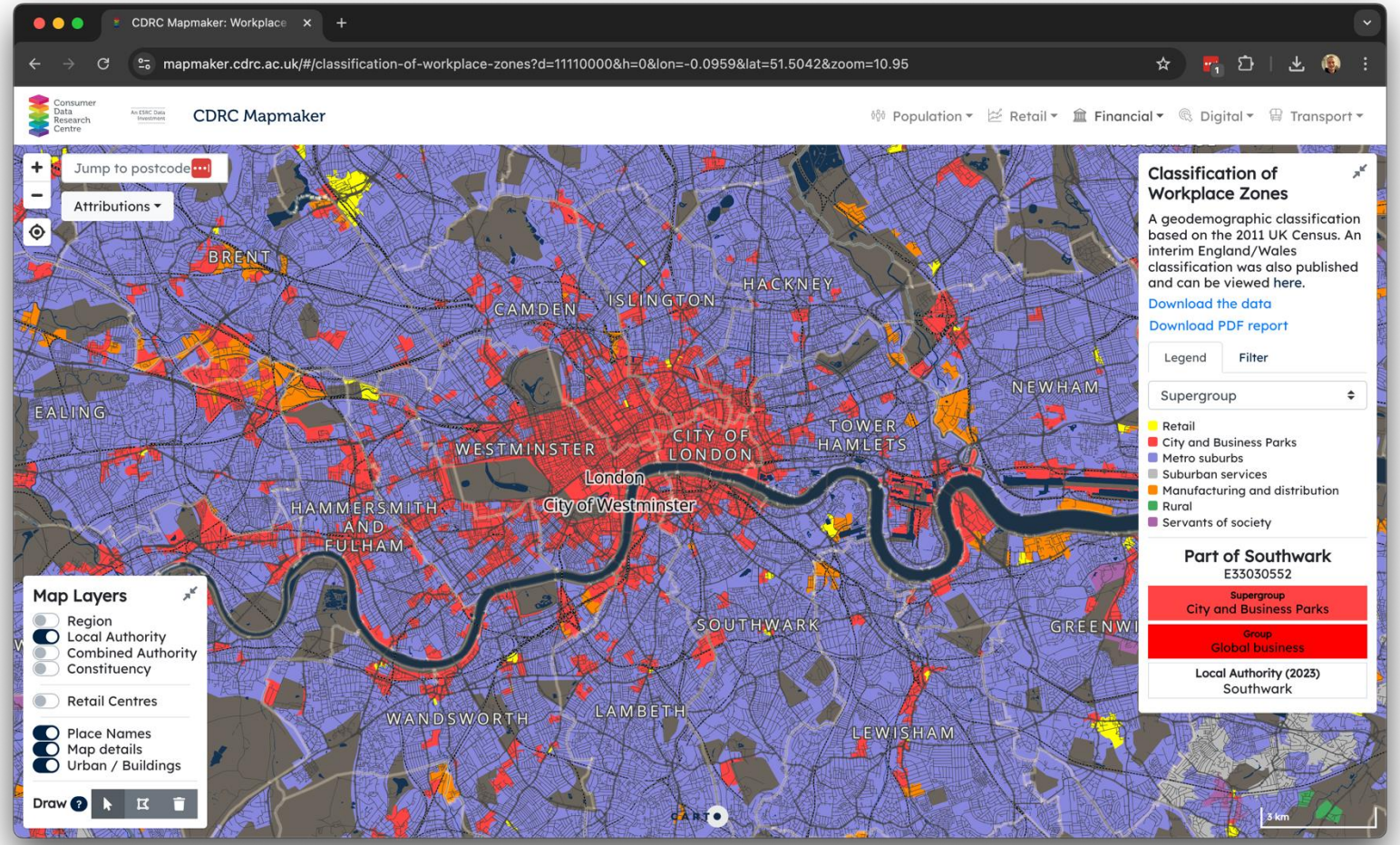
2011 Output Area Classification on mapmaker.geods.ac.uk

Geodemographics



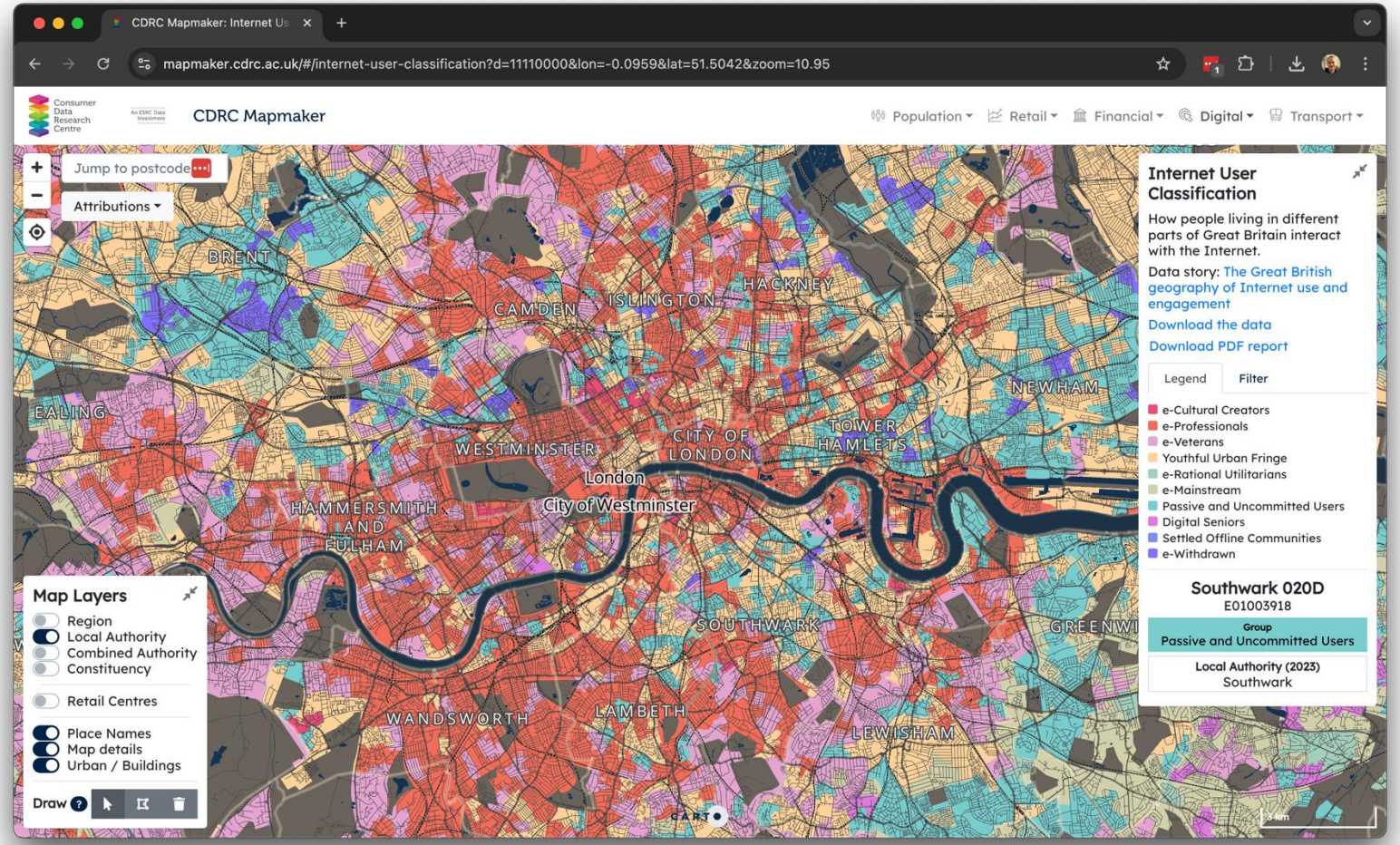
2021 London Output Area Classification on mapmaker.geods.ac.uk

Geodemographics



Workplace Zone Classification on mapmaker.geods.ac.uk

Geodemographics



Internet User Classification on mapmaker.geods.ac.uk

Limitations

- Highly dependent on the input data.
- Input data can get old very quickly (depending on the topic).
- Inherent biases within the input data (see Dalton and Thatcher 2015).

Applications

Using the geodemographic classification as input for further analysis:

- Harris *et al*. 2007: differences in school choice between social groups
- Brunsdon *et al*. 2011: participation in higher education
- Martin *et al*. 2018: analysis of travel-to-work flows
- Goodman *et al*. 2011: socio-economic inequalities in exposure to air pollution
- Trasberg and Cheshire 2021: profiling changing activity patterns

Internet user classification

Singleton *et al*. 2020:

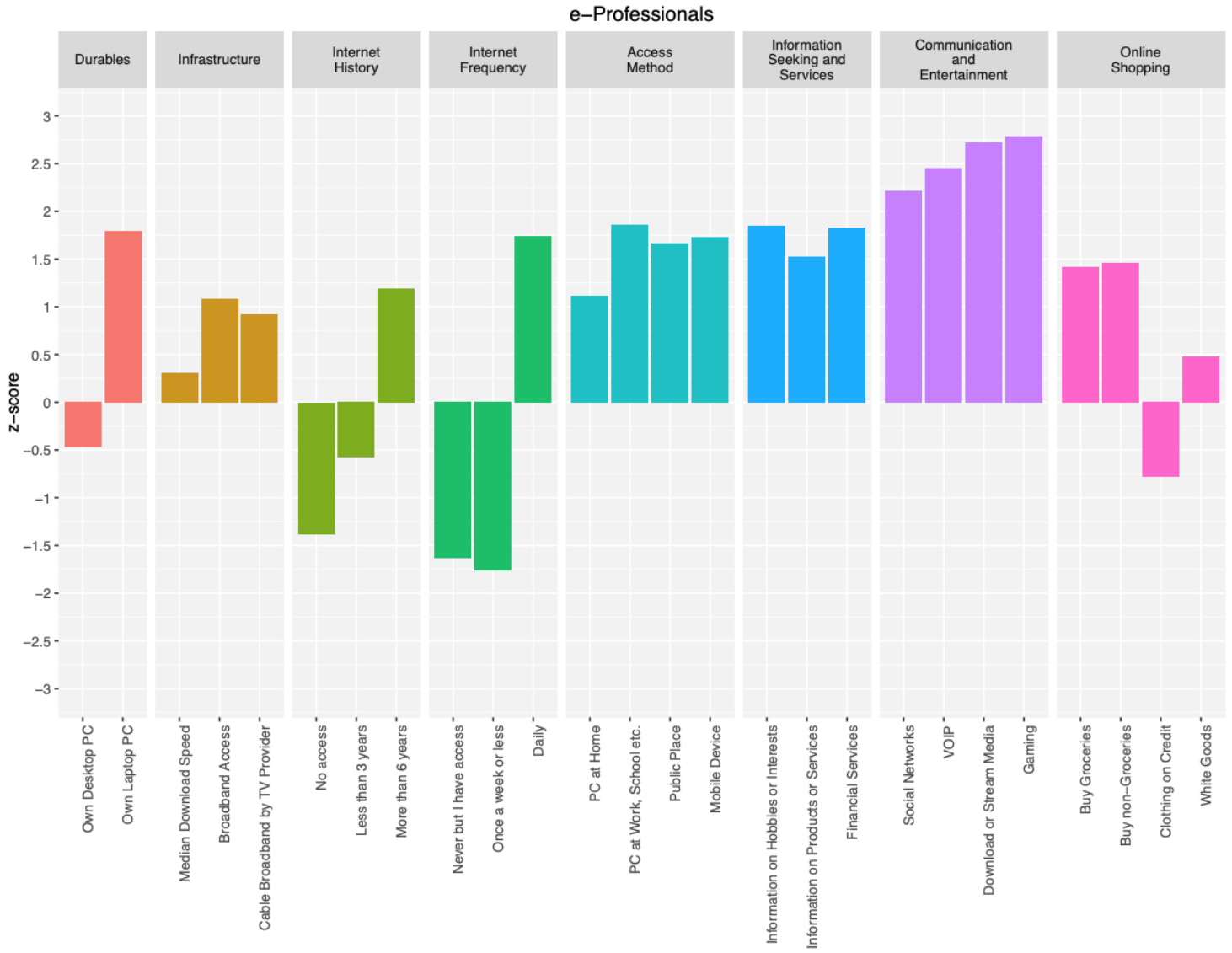
- Bespoke classification created by GeoDS.
- ‘Profiles of internet use and engagement’.
- Built from a range of consumer data, survey data, and open data.

Internet user classification

Several noteworthy variables:

- British Population Survey: internet access, frequency of internet usage, access to PC, type of internet use.
- Transactional (consumer) data on online shopping.
- Average broadband speed.
- Census variables such as age, ethnicity.
- National Statistics Socio-economic classification (NS-SEC)

Internet user classification



Internet User Classification mean attributes of the *e–Professionals*

Internet user classification

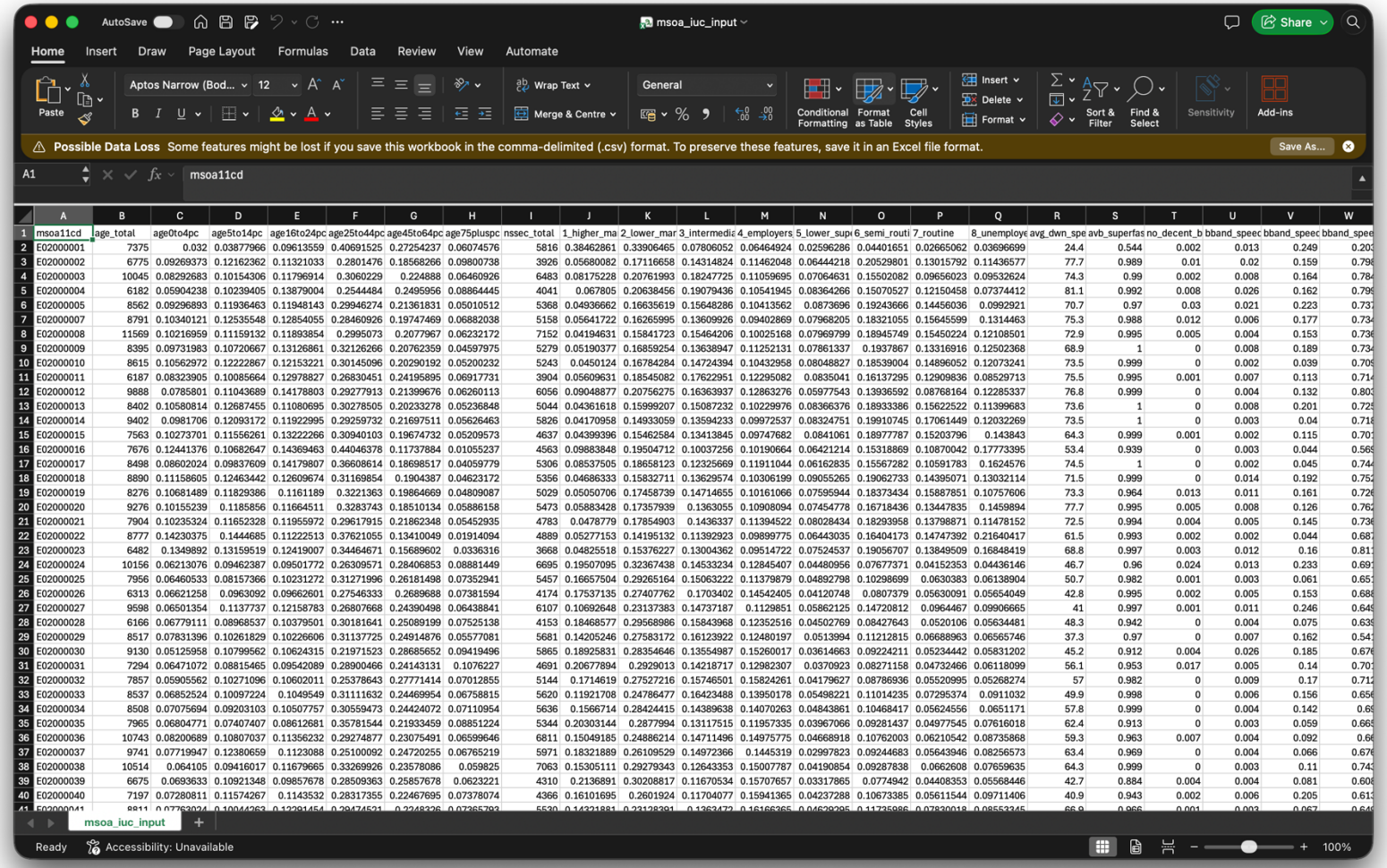
e-Professionals:

“This Group has high levels of Internet engagement, particularly regarding social networks, communication, streaming and gaming, but relatively low levels of online shopping, besides groceries. They are new but very active users, with a very high proportion of the population engaging on a daily basis. (...) Geographically, this Group is mainly located close to the city centre or within the proximity of Higher Education Institutes, where infrastructure accessibility, such as cable broadband, is sufficient.”

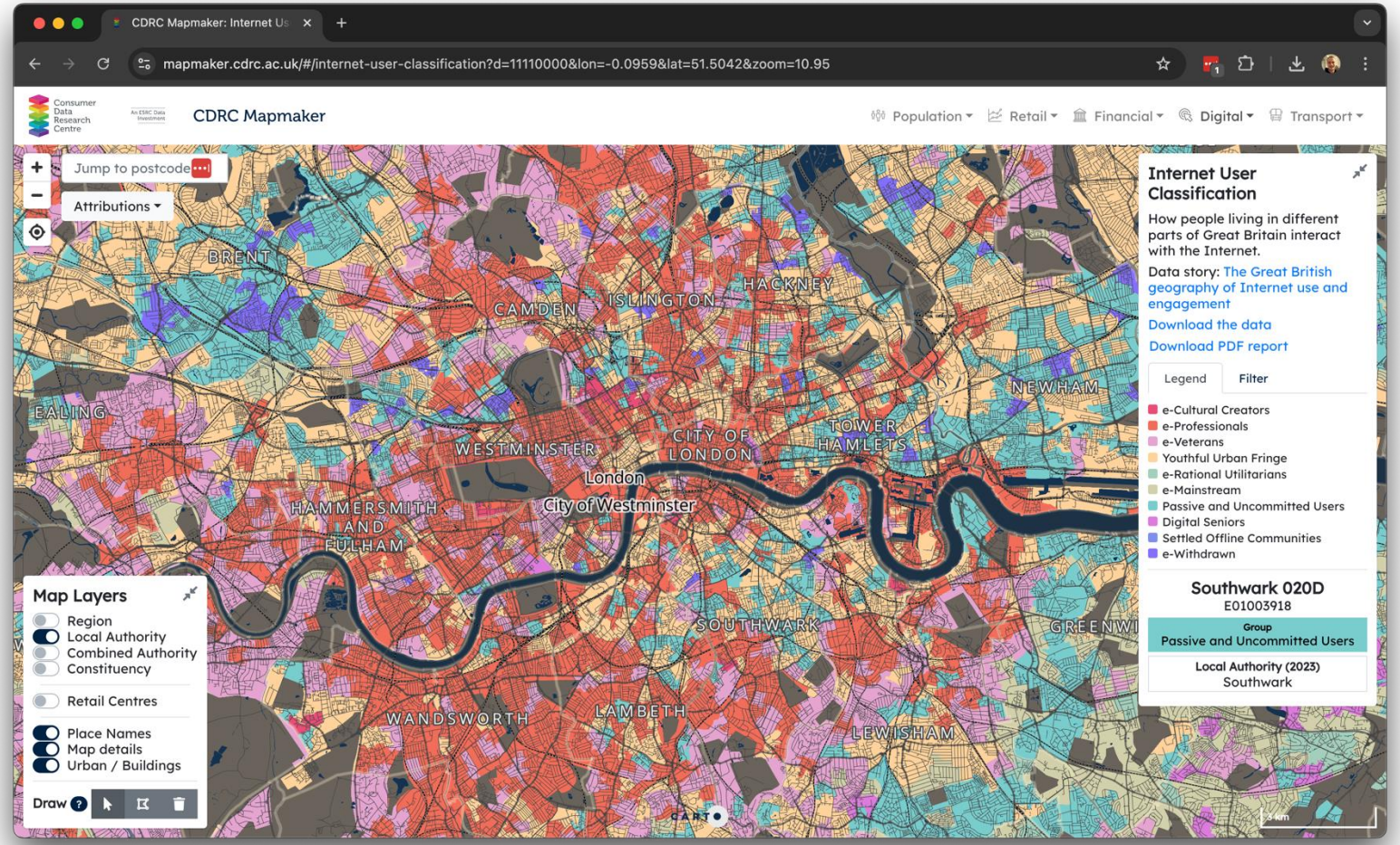
Internet user classification

- Measures of access to and use of internet.
- Identification of areas to target potential interventions.
- Analysis of areas where people are likely to work from home.

Workflow


msoa_iuc_input
Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.
Save As...
A1
msoa11cd
msoa_iuc_input
Ready
Accessibility: Unavailable
100%

Workflow


CDRC Mapmaker: Internet Us
mapmaker.cdrc.ac.uk/#/internet-user-classification?d=11110000&lon=-0.0959&lat=51.5042&zoom=10.95
Consumer Data Research Centre
CDRC Mapmaker
Population
Retail
Financial
Digital
Transport
Jump to postcode
Attributions
BRENT
CAMDEN
ISLINGTON
HACKNEY
EALING
NEWHAM
WESTMINSTER
CITY OF LONDON
TOWER HAMLETS
London
City of Westminster
HAMMERSMITH AND FULHAM
SOUTHWARK
WANDSWORTH
LAMBETH
LEWISHAM
Internet User Classification
How people living in different parts of Great Britain interact with the Internet.
Data story: The Great British geography of Internet use and engagement
Download the data
Download PDF report
Legend
Filter
e-Cultural Creators
e-Professionals
e-Veterans
Youthful Urban Fringe
e-Rational Utilitarians
e-Mainstream
Passive and Uncommitted Users
Digital Seniors
Settled Offline Communities
e-Withdrawn
Southwark 020D
E01003918
Group
Passive and Uncommitted Users
Local Authority (2023)
Southwark
Map Layers
Region
Local Authority
Combined Authority
Constituency
Retail Centres
Place Names
Map details
Urban / Buildings
Draw
CARTO
3 km

Workflow

Typical workflow:

1. Choose input domains.
2. Select associated quantifiable variables.
3. Measure variables for multi-collinearity.
4. Transform and standardise variables.
5. Choose a clustering method.
6. Determine the number of clusters.
7. Interpret, test, and finalise findings.

Input domain

- What type of application is your classification aimed at?
- Who or what are you segmenting? Individuals, households, postcodes, output areas, wards?
- What is the scale of data available?
- What difficulties might arise from combining datasets?

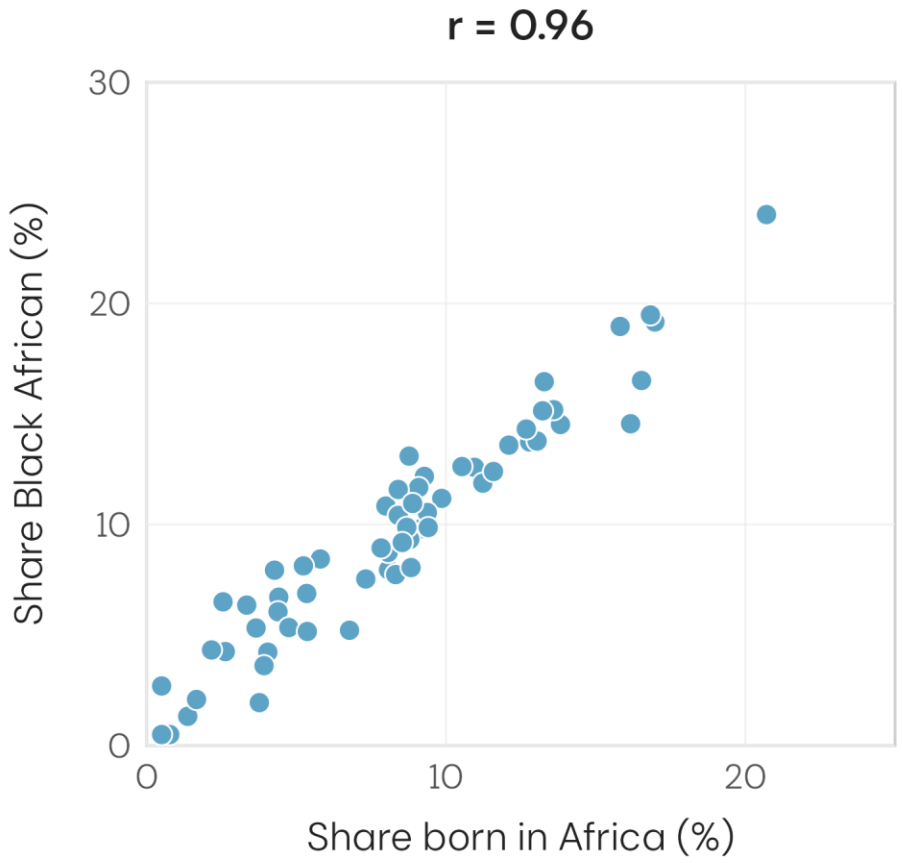
Variables

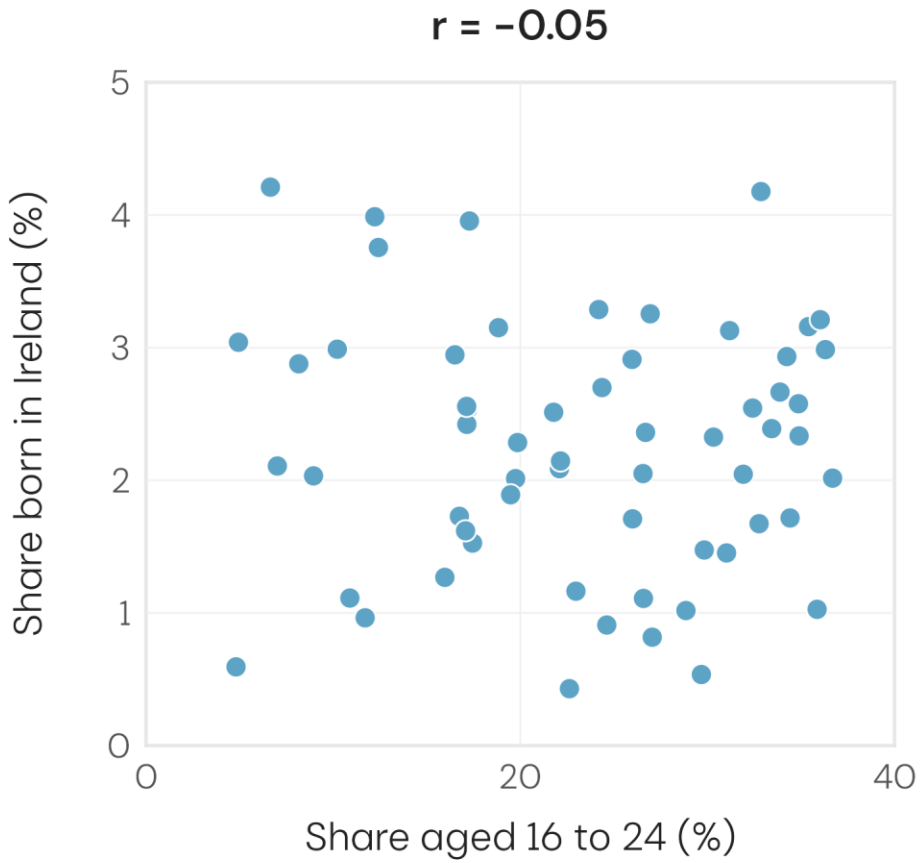
- Partially subjective: no single or unique set of variables.
- Balancing act of theory, available data and statistical considerations.
- Selection of variables directly affects what is classified.

Multi-collinearity

- The Pearson correlation coefficient ranges from −1 to 1.
- A value closer to −1 or 1 indicates a stronger linear association between two variables.
- Rule of thumb: variables with a correlation above about ±0.7 carry nearly the same information.
- You may choose to replace or retain a variable.

Highly correlated variables count twice





Standardise variables

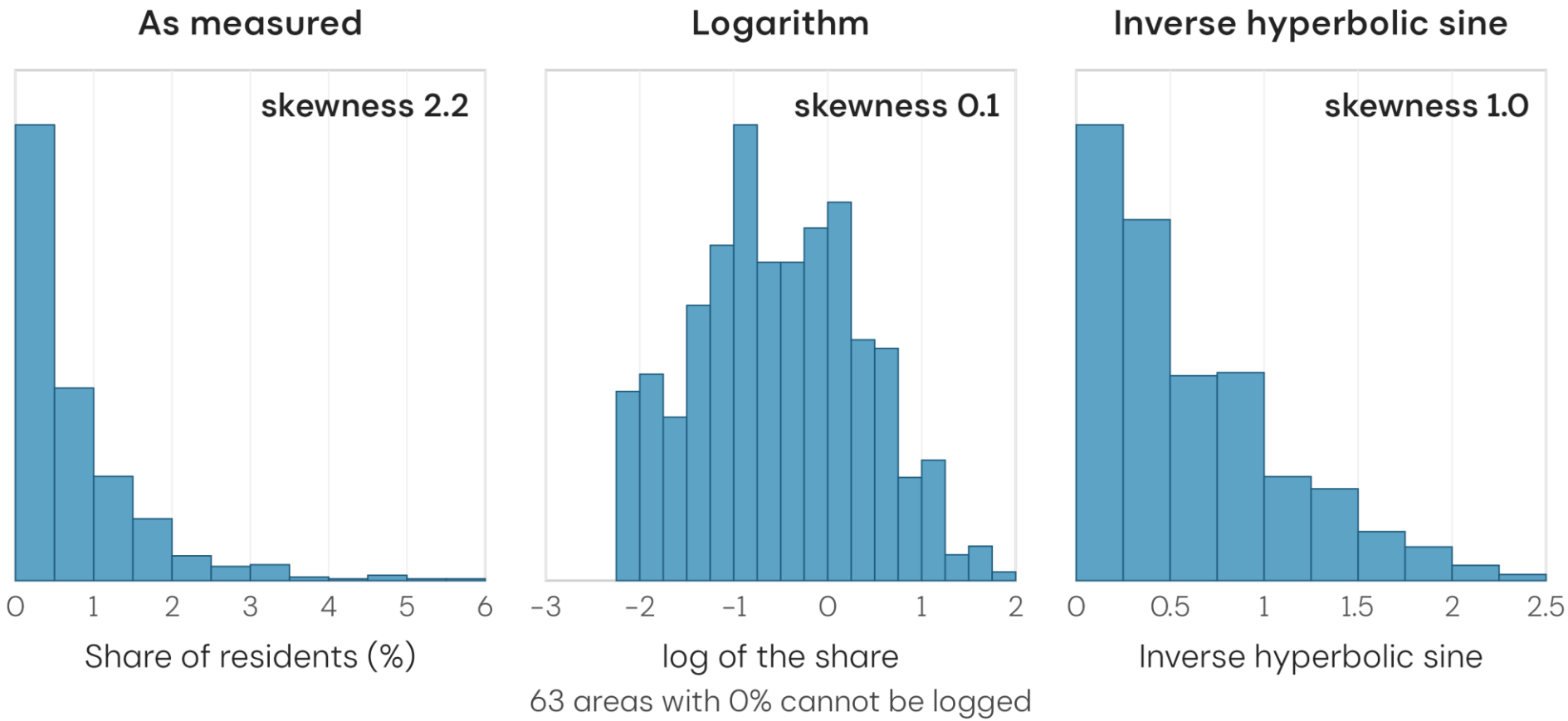
Normalise to control for non-uniformity of areal units:

- Expressing variables as percentages or proportions (e.g. using a relevant population denominator).

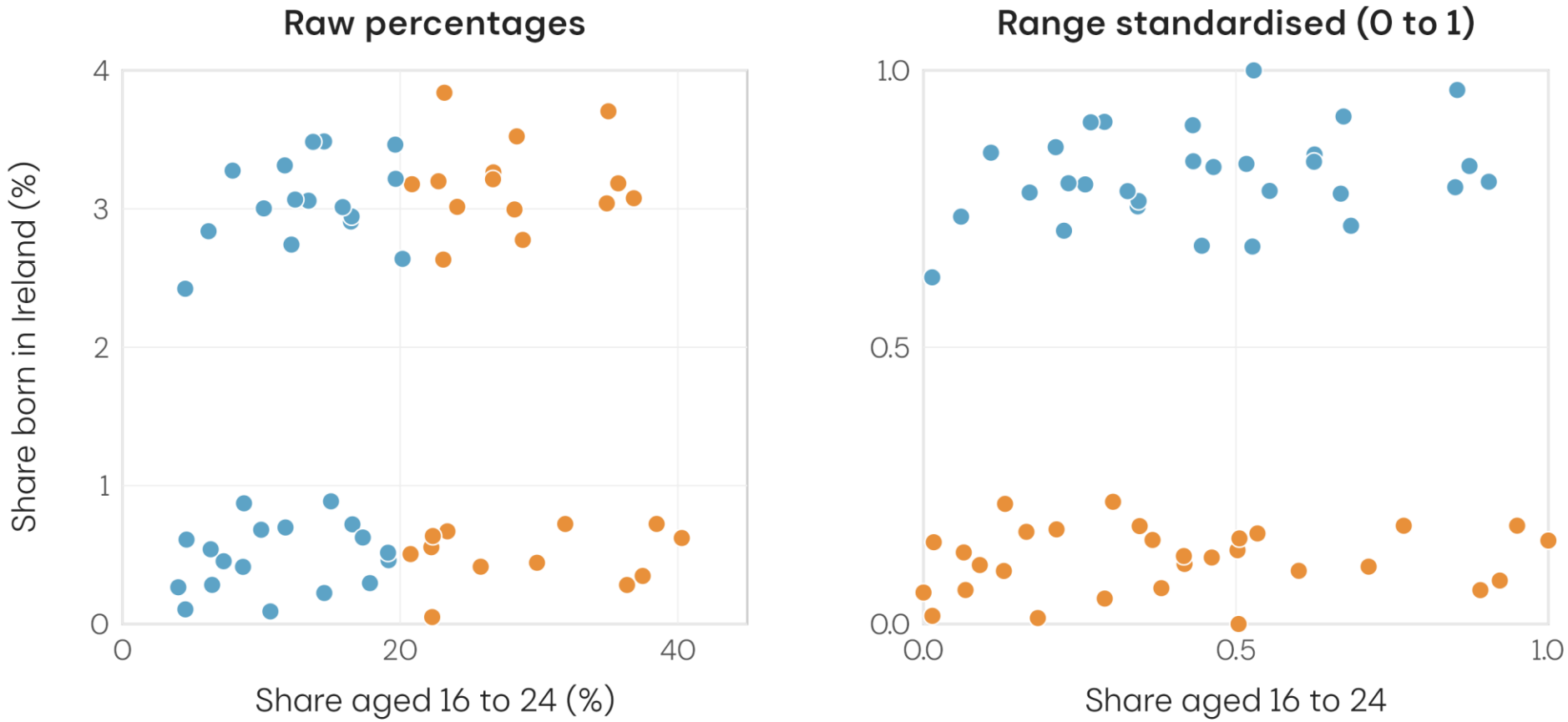
Standardise between variables to account for value ranges:

- Transform all variables to be on the same scale.
- z-scores, range standardisation.

Skewed variables: transform before clustering



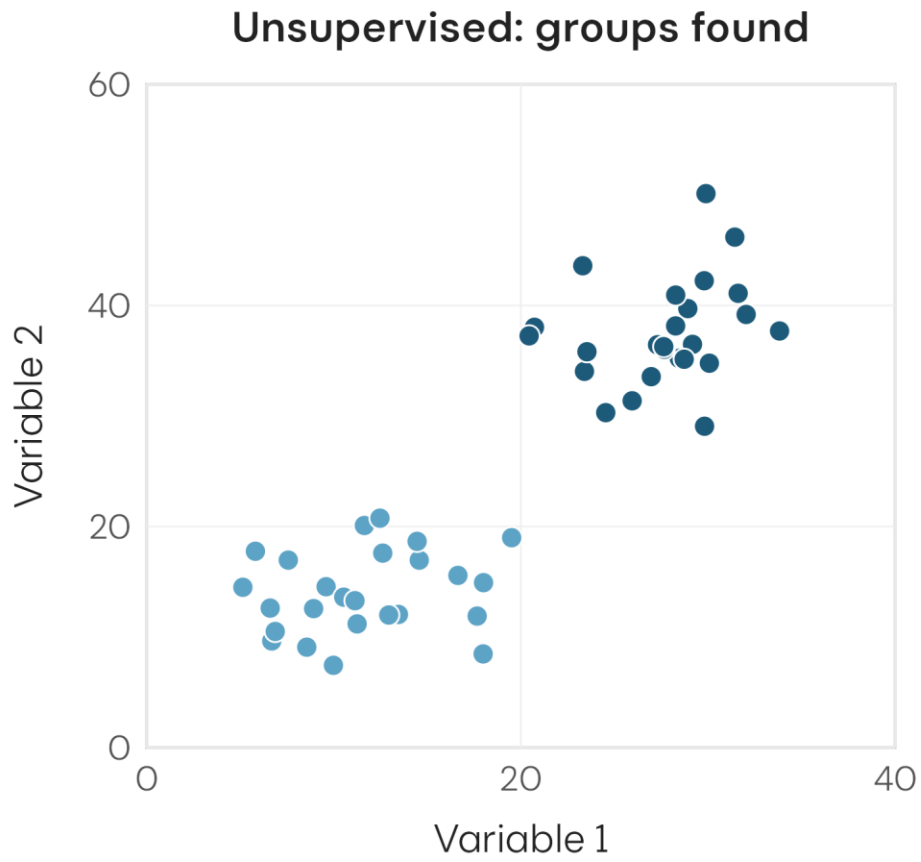
Standardise: otherwise the largest scale decides

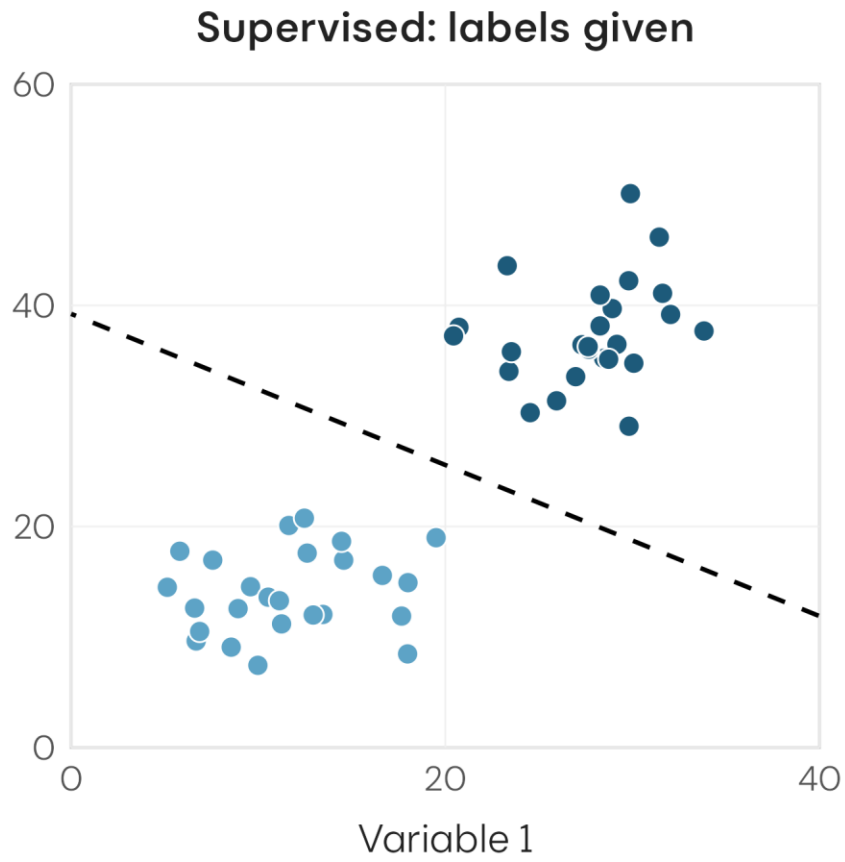


Clustering

- Different methods often create different groups.
- Selecting the best outcome and optimal number of clusters often relies on user intuition and judgment.
- The process can be computationally intensive, depending on the size of the input data.
- The methods employed are unsupervised, meaning they rely on patterns in the data without prior labels.

Unsupervised finds groups, supervised predicts labels

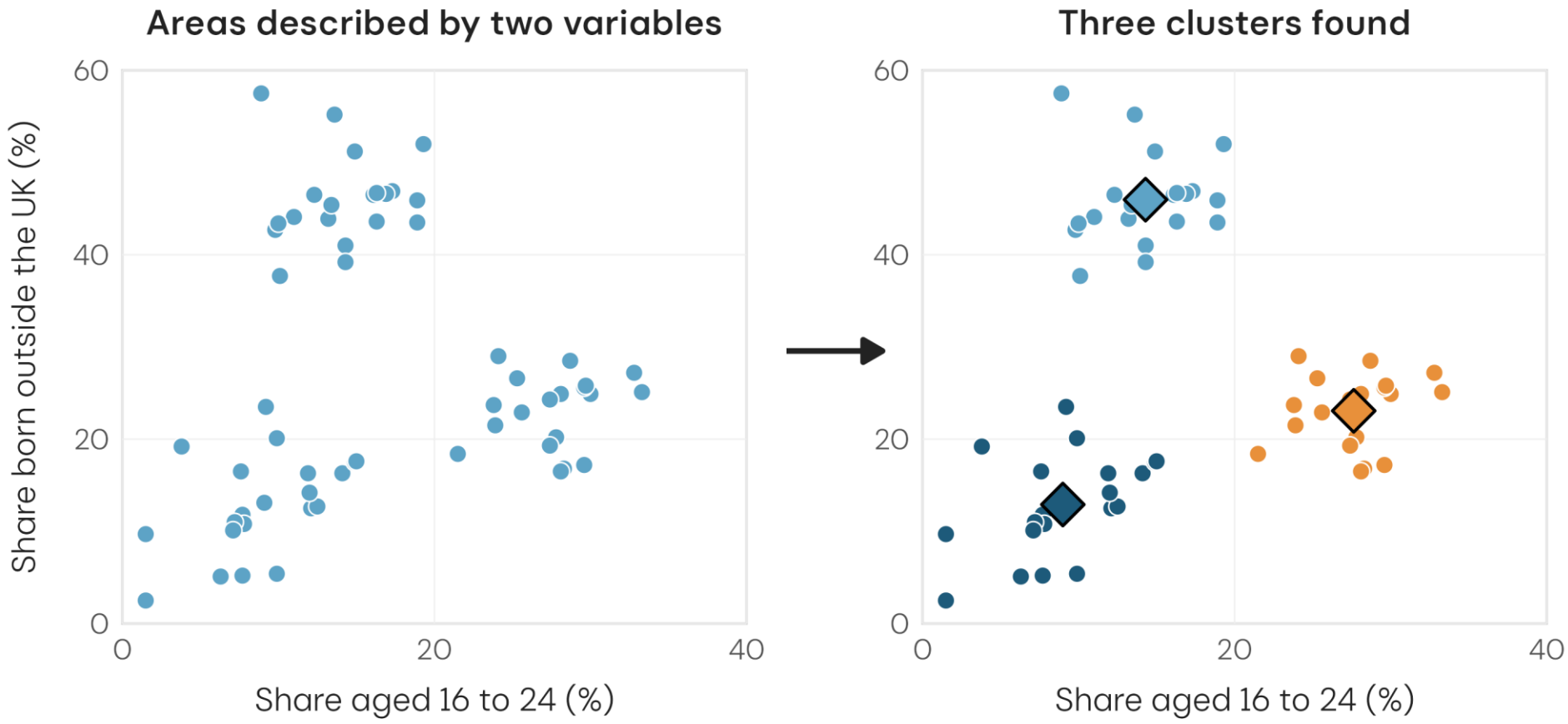




k-means

- Assign geographic areas with common underlying attributes to similar classification groups.
- `k` clusters (pre-defined) of `n` individual observations.
- Each observation can have any number of attribute data.
- Used in: Internet User Classification, ONS 2011 Output Area Classification, ONS 2021 Output Area Classification, Night-Worker classification.

k-means groups areas that are similar



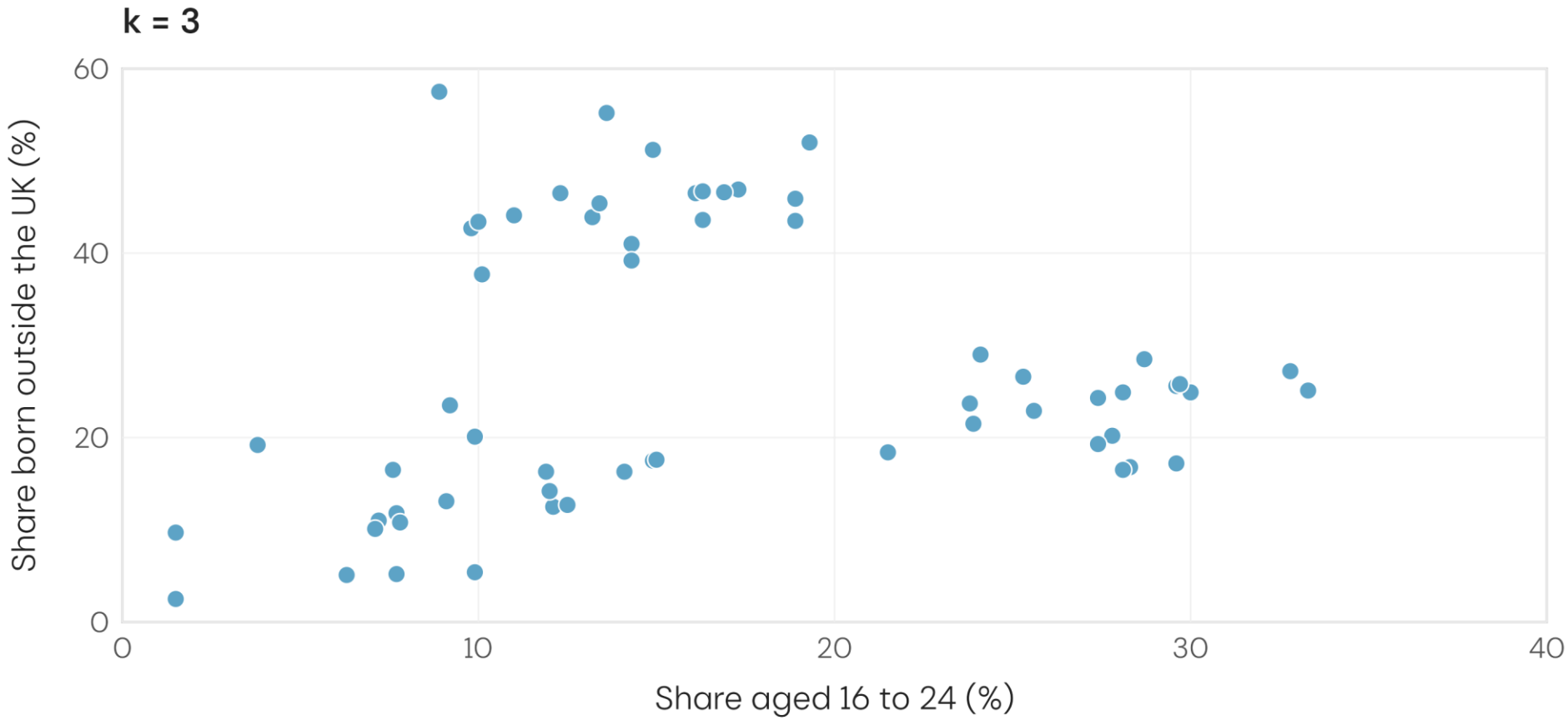
k-means

- Minimising the distance between an observation's input variables to the means of the respective cluster groups.
- Maximising the distance between cluster groups.
- Number of clusters defined a priori.

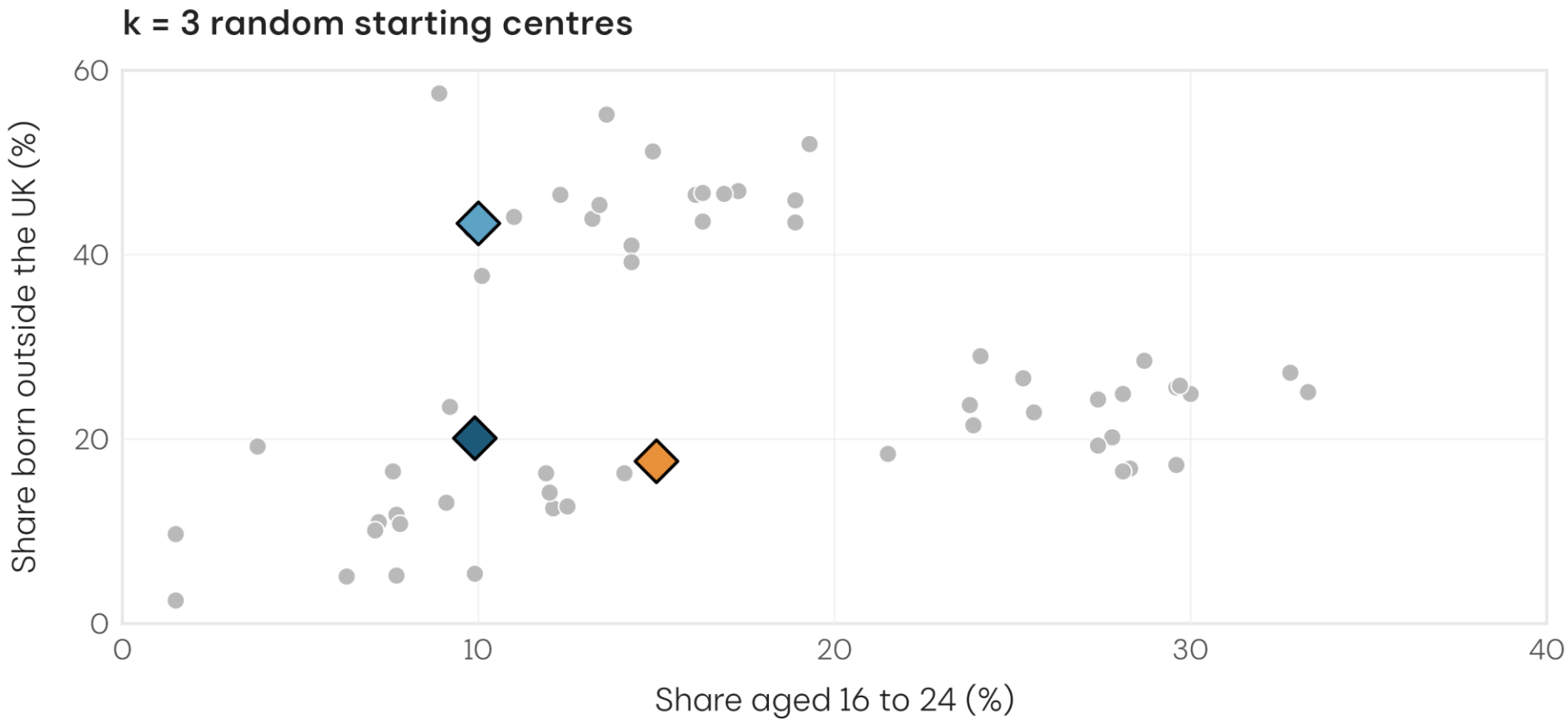
Process

- Step 1: identify your k
- Step 2: randomly identify k distinct data points as initial cluster centre
- Step 3: assign each observation to the nearest cluster
- Step 4: calculate the mean of each cluster
- Step 5: repeat with mean value becoming new cluster centre until no change

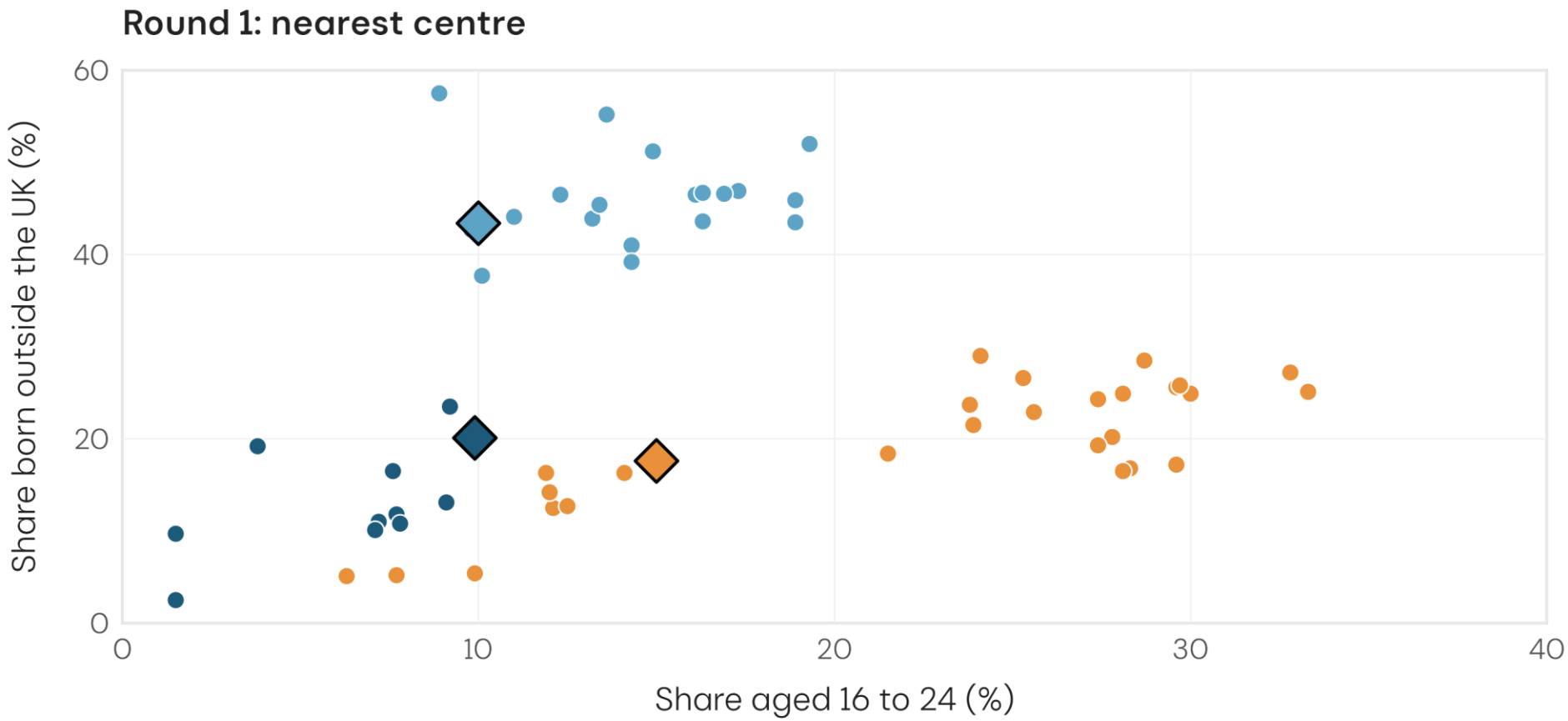
Step 1: choose k



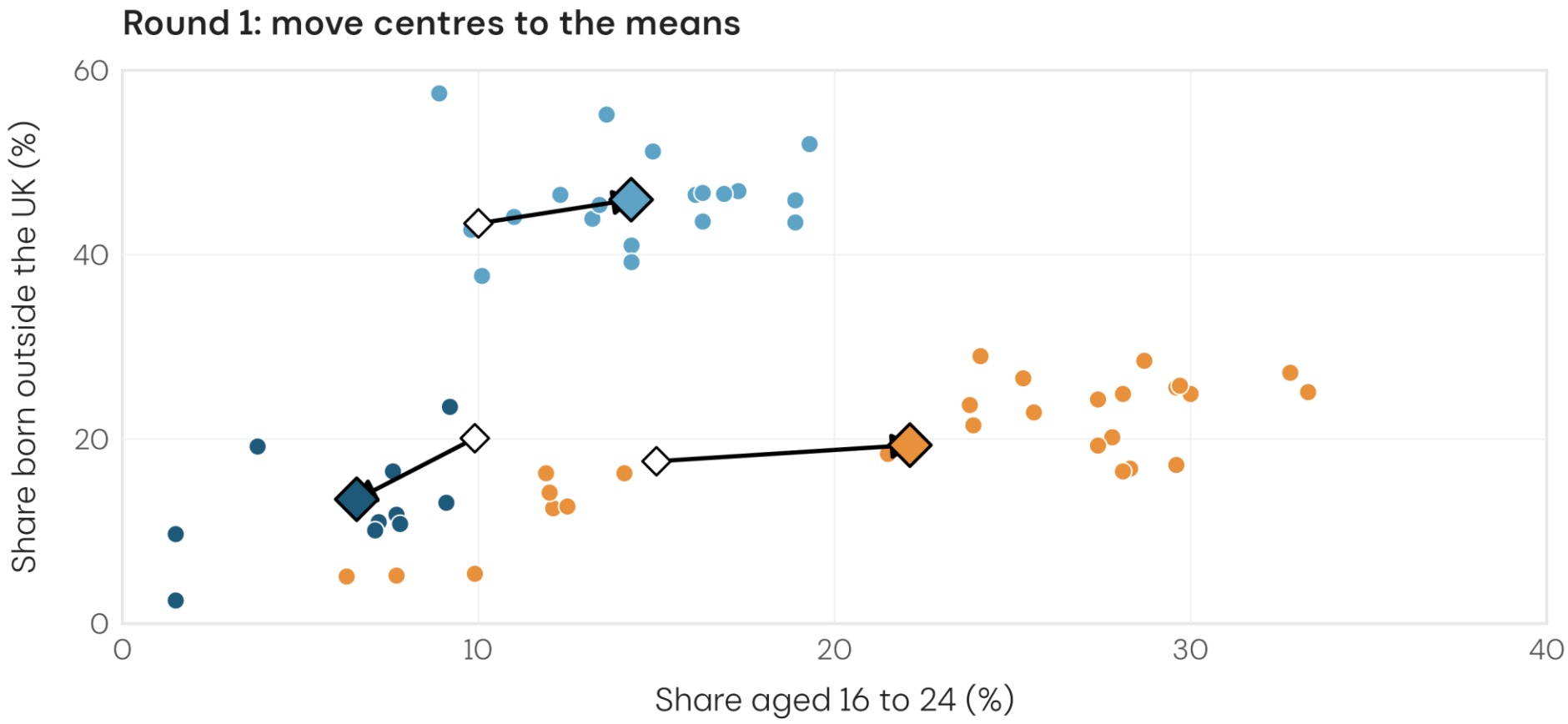
Step 2: place k random centres



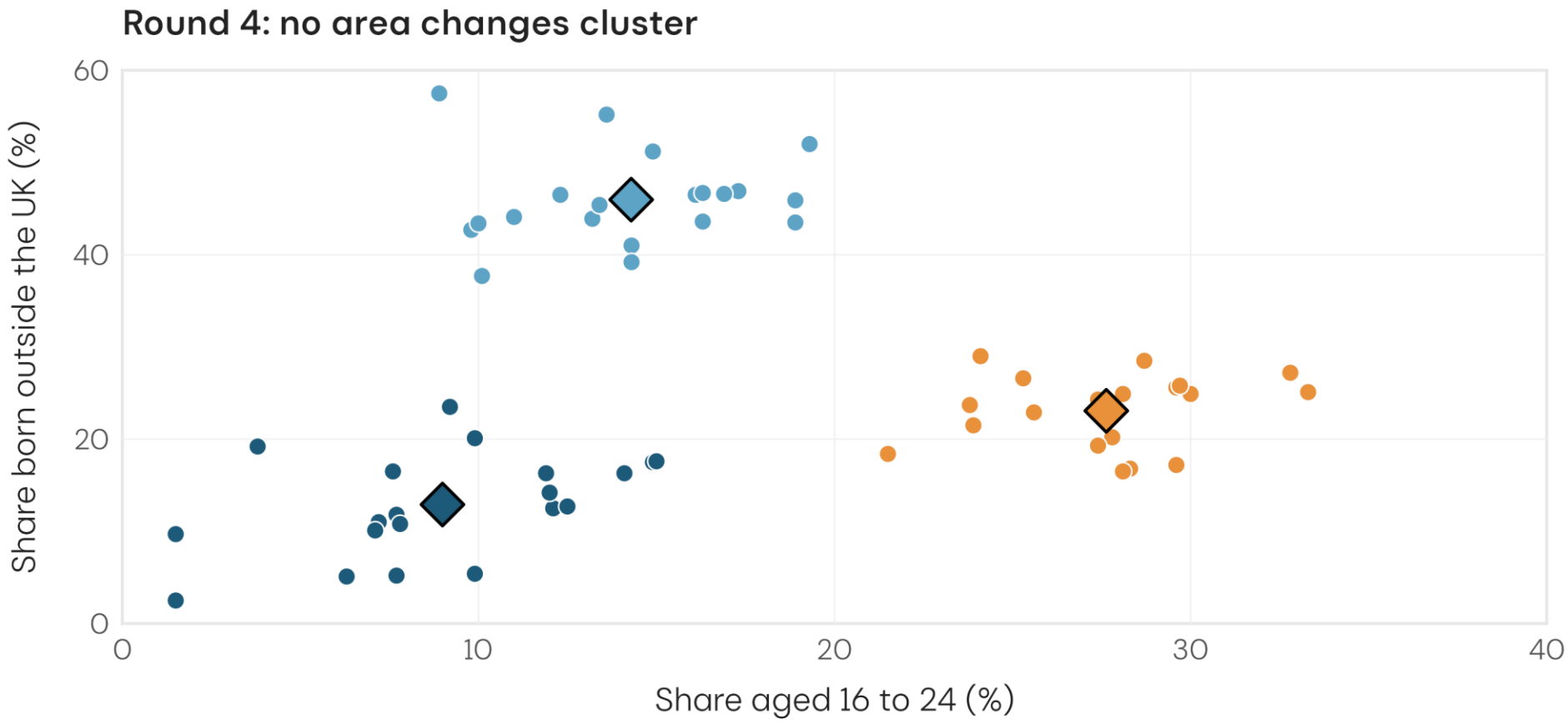
Step 3: assign each area to the nearest centre



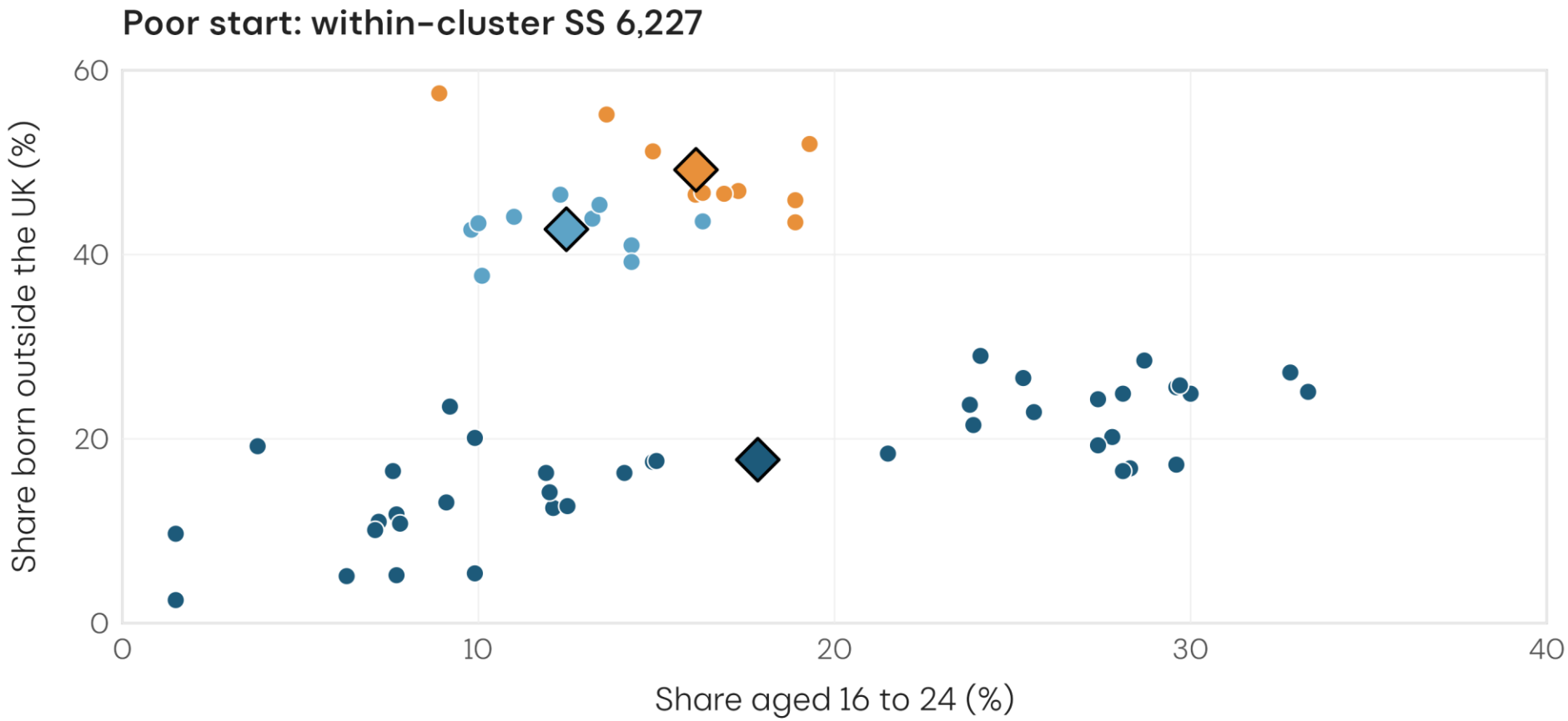
Step 4: move each centre to the mean of its cluster



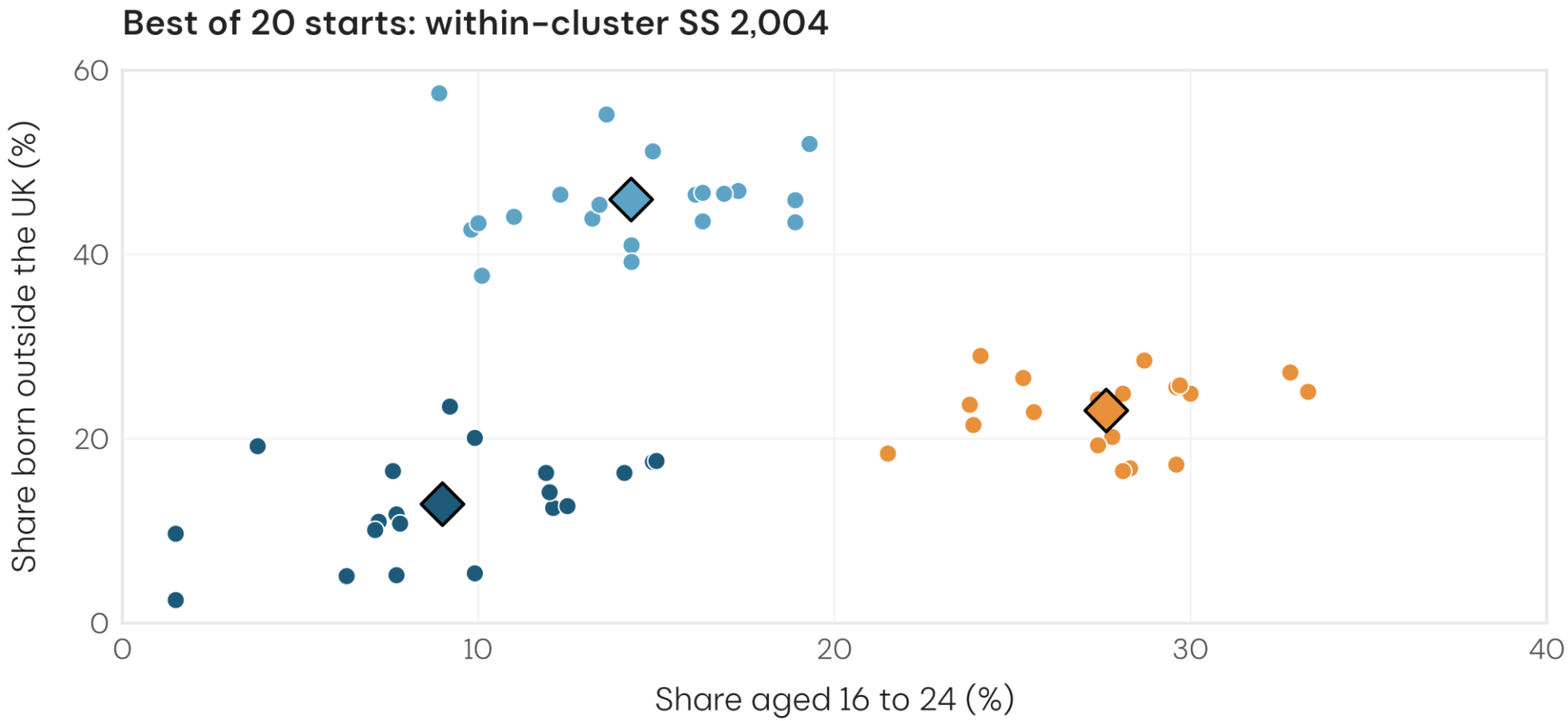
Step 5: repeat until nothing changes



A poor random start can give a poor result



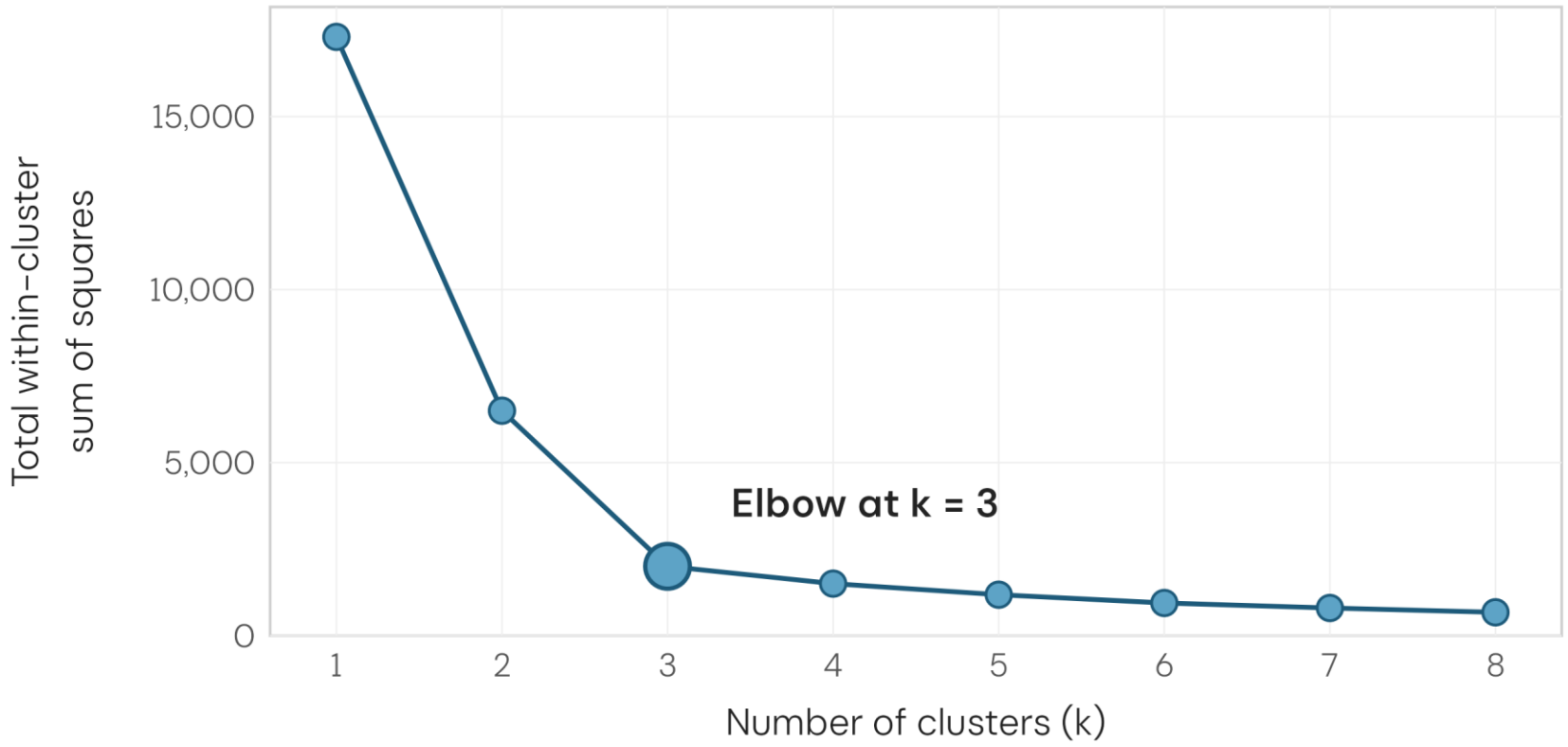
Run many random starts and keep the best



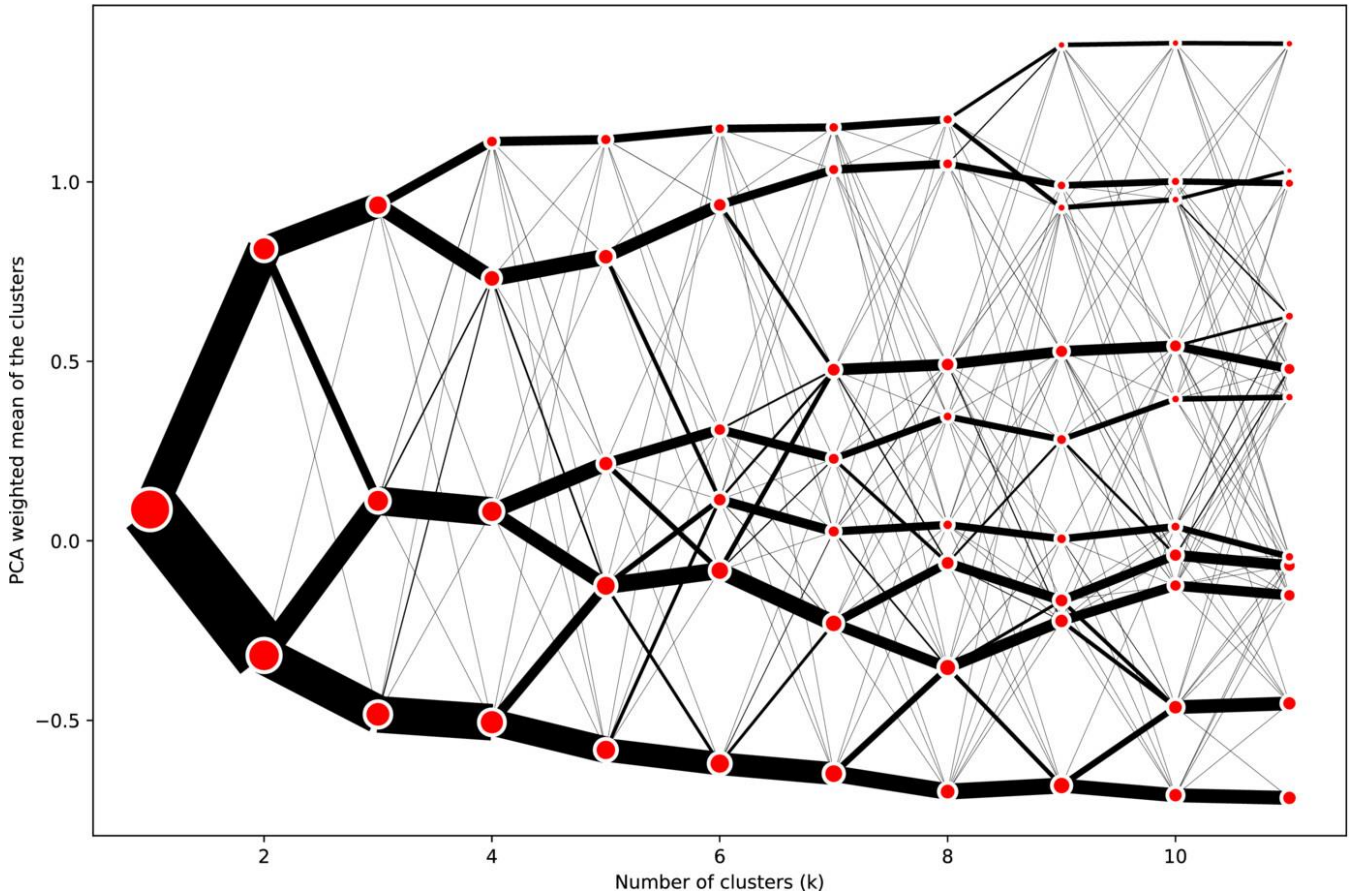
Number of clusters

- Too few: too much variation within the groups.
- Too many: overfitting and splitting similar observations.

The elbow: where extra clusters stop helping



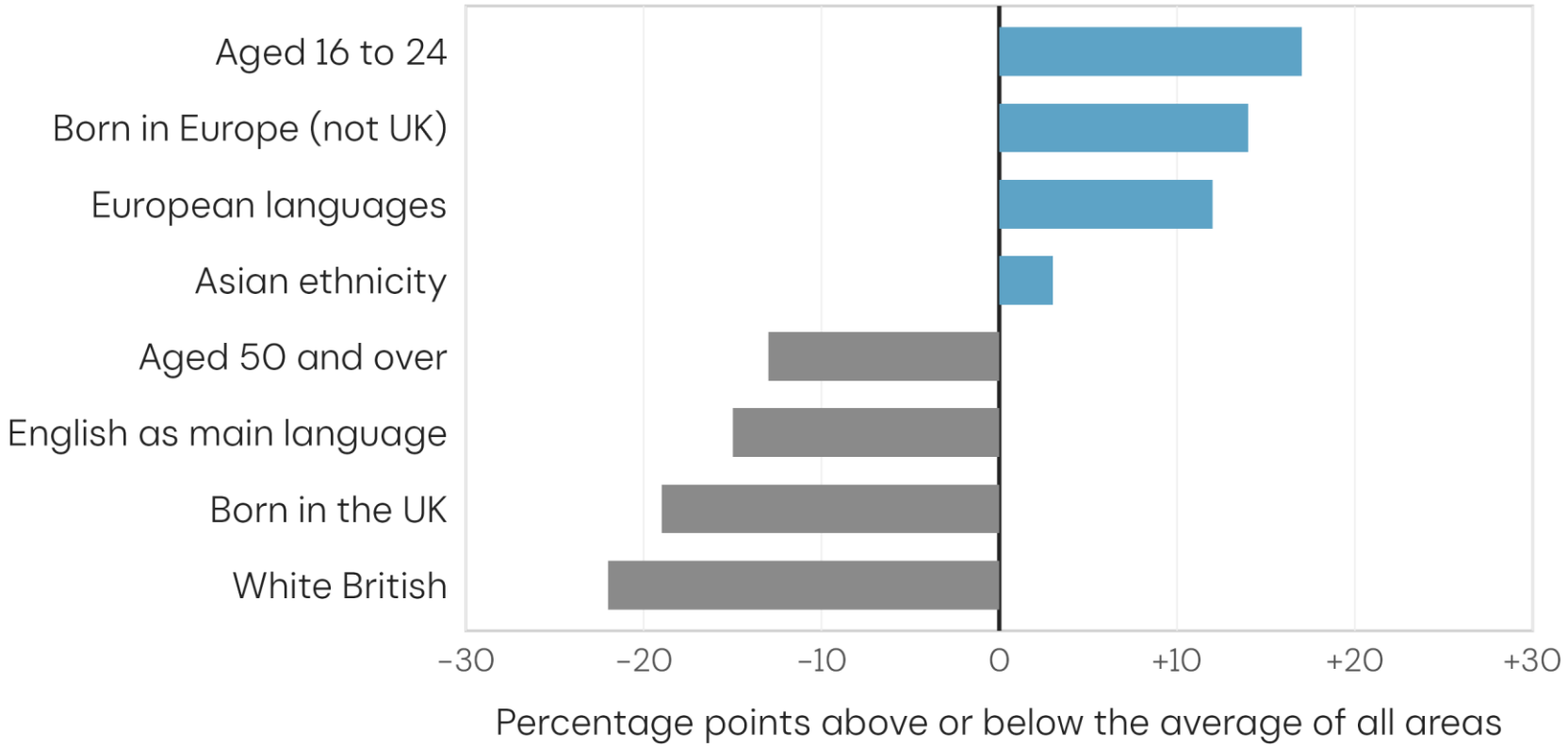
Clustergram: how areas move as k increases



Interpretation of clusters

- Look at the means of the input data for each cluster (‘signature’).
- Can use spider or bar plots to characterise the cluster.
- Not spatial in nature (different from DBSCAN).

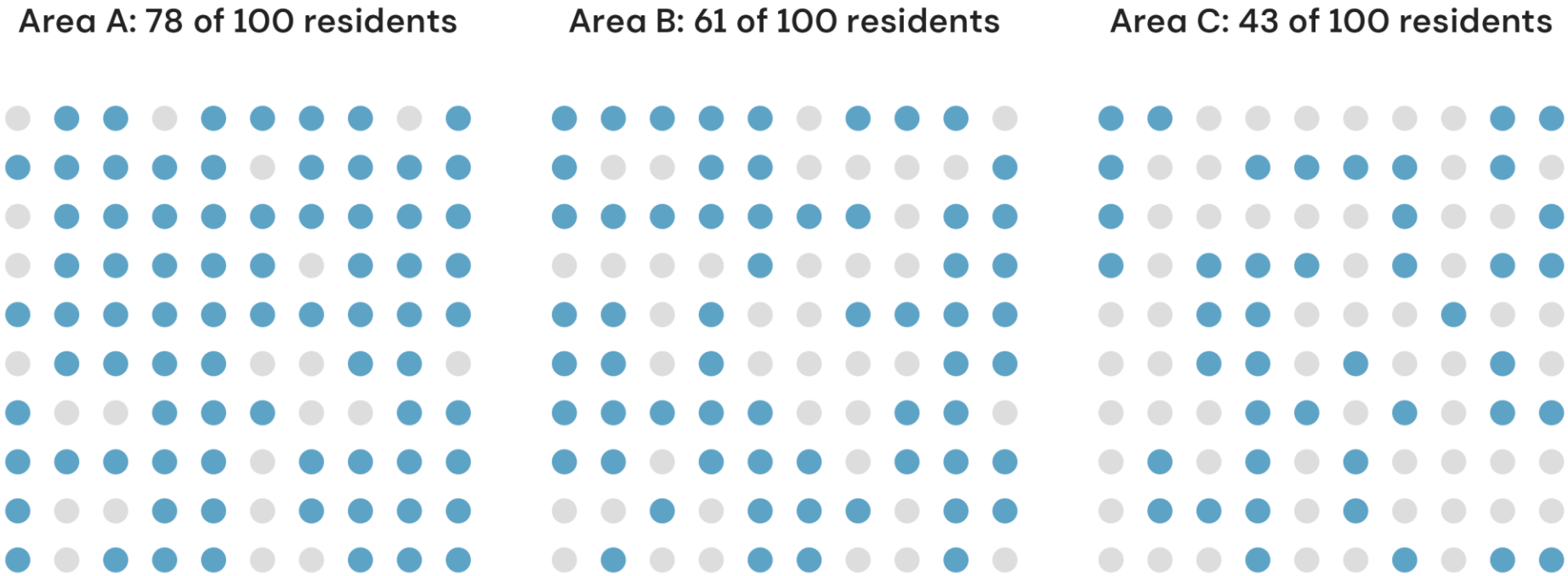
A cluster profile shows what stands out



Cluster labels

- Should generalise the characteristics of each group.
- Should be clear for users unfamiliar with geodemographics.
- Should not draw on assumptions that cannot be applied by the data.
- Be aware of the ecological fallacy: a label describes the area, not every resident.
- Create pen portraits for easy interpretation.

A cluster label describes the area, not every resident



Conclusion

- Geodemographics as the analysis of people by where they live.
- Typically, a form of unsupervised machine learning is used.
- Once created: benchmarking against additional quantitative data.
- Sometimes the spatial scale should be re-considered.
- Recent classifications: London Output Area Classification 2021, Output Area Classification England and Wales 2021, UK Output Area Classification 2021, London Night-workers classification.

Questions

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